

K-Medoids Clustering



What is K-Medoids?

K-Medoids is an **unsupervised machine learning clustering algorithm** similar to K-Means. It groups data into clusters using **actual data points** as cluster centers instead of mean values. These center points are called **Medoids**.

Main Idea of K-Medoids

- The algorithm:
1. Selects K medoids randomly
 2. Assigns points to nearest medoid
 3. Replaces medoids to reduce total distance
 4. Repeats until clusters stabilize

Important Terms

Term	Meaning
Cluster	Group of similar data points
Medoid	Actual data point representing a cluster
K	Number of clusters

Difference Between K-Means and K-Medoids

Feature	K-Means	K-Medoids
Center	Mean value	Actual data point
Sensitive to Outliers	Yes	No
Accuracy	Medium	Better with noisy data
Speed	Faster	Slower

Distance Formula

K-Medoids usually uses Euclidean Distance.

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Where:

Symbol	Meaning
(x_1, y_1)	First point coordinates
(x_2, y_2)	Second point coordinates
(d)	Distance between points

Working of K-Medoids

Steps

1. Select K random medoids
2. Assign each point to nearest medoid
3. Compute total cost
4. Swap medoid with non-medoid point

5. If cost decreases, update medoid
 6. Repeat until no improvement occurs
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Example of K-Medoids



Suppose we have customer data:

Customer	Spending Score
A	High
B	High
C	Low
D	Low

If ($K = 2$):

- Cluster 1 → High spenders
- Cluster 2 → Low spenders

The medoid is the actual customer point nearest to the cluster center.

Cost Function

K-Medoids minimizes total dissimilarity:

$$Cost = \sum_{i=1}^n distance(point_i, medoid)$$

PAM Algorithm

K-Medoids commonly uses:

PAM (Partitioning Around Medoids)

PAM works by:

- Selecting medoids
 - Swapping medoids
 - Minimizing total clustering cost
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Advantages of K-Medoids

- Robust to outliers
 - Works well with noisy data
 - Uses real data points as centers
 - More stable than K-Means
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Disadvantages of K-Medoids

- Slower than K-Means
 - Computationally expensive
 - Not suitable for very large datasets
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Applications of K-Medoids

- Customer Segmentation
 - Fraud Detection
 - Medical Data Analysis
 - Bioinformatics
 - Recommendation Systems
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Comparison with Other Algorithms

Algorithm	Type	Outlier Handling	Speed
K-Means	Unsupervised	Poor	Fast
K-Medoids	Unsupervised	Good	Slow
SVM	Supervised	Medium	Medium
Naive Bayes	Supervised	Medium	Very Fast

Python Example

```
from sklearn_extra.cluster import KMedoids
import numpy as np

# Data
X = np.array([
    [1, 2],
    [1, 4],
    [1, 0],
    [10, 2],
    [10, 4],
    [10, 0]
])

# Create model
kmedoids = KMedoids(n_clusters=2)

# Train model
kmedoids.fit(X)

# Cluster labels
print(kmedoids.labels_)
```

Important Interview Questions

1. What is K-Medoids?

An unsupervised clustering algorithm using actual data points as cluster centers.

2. What is a Medoid?

A real data point representing the center of a cluster.

3. Difference between K-Means and K-Medoids?

K-Means uses mean values, while K-Medoids uses actual data points.

4. Why is K-Medoids better for noisy data?

Because it is less sensitive to outliers.

Short Notes

Definition

K-Medoids is an unsupervised clustering algorithm that groups data using actual data points as cluster centers.

Main Goal

Minimize total distance between points and medoids.

Advantages

- Robust to outliers
- Stable clustering

Disadvantages

- Slow for large datasets

- Computationally expensive

Applications

- Fraud detection
- Customer segmentation
- Medical analysis